

Back of the Envelope Estimation

To estimate the scale of the autocomplete system, we make the following assumptions.

Assumptions

- **10 million Daily Active Users (DAU)**
- Each user performs **10 searches per day**
- Each query string consumes **20 bytes**

Query size calculation

Assume:

- 1 character = 1 byte (ASCII encoding)
- Each query contains:
 - 4 words
 - 5 characters per word on average

So:

$4 * 5 = 20$ bytes per query

Autocomplete requests generated

For every character typed, the client sends a request to the backend for suggestions.

Example while typing "dinner":

None

`search?q=d`

`search?q=di`

`search?q=din`

`search?q=dinn`

```
search?q=dinne
```

```
search?q=dinner
```

Thus, on average:

- **20 requests** are generated per search query.

Query Per Second (QPS)

Total requests per day:

10,000,000 users * 10 queries/day * 20 characters = 2,000,000,000 requests/day

Convert to QPS:

$$\text{QPS} = (10,000,000 * 10 * 20) / (24 * 3600) \approx 24,000$$

So:

- **Average QPS $\approx$ 24,000**

Peak QPS

Assume peak traffic is 2 times the average traffic:

$$\text{Peak QPS} = 24,000 * 2 \approx 48,000$$

So:

- **Peak QPS $\approx$ 48,000**

New data added daily

Assume:

- 20% of queries are new queries.

Daily storage requirement:

$$10,000,000 * 10 * 20 * 20\% = 0.4 \text{ GB}$$

So:

- **0.4 GB of new query data is added every day**

Final Estimation Summary

Metric	Value
Daily Active Users	10 Million
Searches per user/day	10
Avg query size	20 bytes
Requests per search	20
Average QPS	~24,000
Peak QPS	~48,000
New storage/day	~0.4 GB

